

INDUSTRIAL TRAINING REPORT
CIVICPULSE — BBMP GRIEVANCE INTELLIGENCE
Ward-wise Civic Demand, Grievance Trends & Municipal Decision Analytics

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DECLARATION

I hereby declare that the Industrial Training Report entitled “CivicPulse — BBMP Grievance Intelligence: Ward-wise Civic Demand, Grievance Trends & Municipal Decision Analytics” is an authentic record of my own work as requirements of Industrial Training during the period from 17 August 2026 to 30 September 2026 for the award of degree of B.Tech. (CSE-DS), ABES Engineering College, Ghaziabad, under the guidance of

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(Signature of student)

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CERTIFICATE

This is to certify that Mr. / Ms. Deepak Dubey has partially completed / completed Industrial Training during the period from 17 August 2026 to 30 September 2026 in AICTE – IBM SkillsBuild Data Analytics with AI Internship Program 2026 / BharatCares as a Partial Fulfillment of Degree of Bachelor of Technology in CSE-DS. He / She was trained in the field of Data Analytics with AI and civic intelligence.

Signature & Seal of Training Manager

Certificate wording / signatory: Under Process — Leave Blank for now

Note: This certificate must be typed on the company letter head.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to AICTE, IBM SkillsBuild, and BharatCares for providing me the opportunity to pursue the Data Analytics with AI Internship Program 2026. I am thankful to the faculty and mentors of ABES Engineering College for their academic support. I also acknowledge the support of the internship learning environment, dataset sources, and open-source analytical tools used in the development of CivicPulse.

The project enabled me to apply data collection, data cleaning, exploratory analysis, machine learning, dashboard development, and technical documentation to a real-world civic problem. I am grateful to everyone who contributed directly or indirectly to the completion of this training project.

Deepak Dubey

ABOUT COMPANY / INDUSTRY / INSTITUTE

The industrial training was completed under the AICTE – IBM SkillsBuild Data Analytics with AI Internship Program 2026, in association with BharatCares. The program is focused on developing practical skills in data analytics with artificial intelligence, progressing from data collection and preparation to information, insight, decision, and action.

The internship context encouraged the development of a meaningful analytical project rather than the addition of machine learning for its own sake. CivicPulse was designed within this framework to analyse historical BBMP grievance records and provide decision-support insights for ward-wise municipal planning.

ABES Engineering College, Ghaziabad, provided the academic context for the industrial training and the Bachelor of Technology programme in Computer Science and Engineering (Data Science).

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ABBREVIATIONS AND NOMENCLATURE

BBMP — Bruhat Bengaluru Mahanagara Palike
KPI — Key Performance Indicator
ML — Machine Learning
EDA — Exploratory Data Analysis
PCA — Principal Component Analysis
CSV — Comma-Separated Values

CHAPTER 1

INTRODUCTION TO PROJECT

CivicPulse — BBMP Grievance Intelligence is a data analytics and decision-support project for Bengaluru, Karnataka. It analyses historical BBMP grievance records from 2020 to 2024 to understand ward-wise civic demand, major complaint categories, temporal changes, and grievance status patterns.

Municipal grievance datasets contain useful evidence about recurring service issues, but raw complaint records are difficult to interpret directly. The project therefore follows the analytical workflow Data → Information → Insights → Decision → Action. It transforms complaint records into measurable indicators and visual outputs that can support municipal planning.

The project is deliberately distinct from a citizen-level complaint triage system. Its focus is historical and strategic: identifying high-volume categories, high-demand wards, persistent patterns, and ward civic profiles. It does not introduce image classification, chatbot, RAG, or NLP complaint classification.

The primary questions addressed are: What are the major civic problems? How does demand change over time? Where is demand concentrated? Which categories are associated with particular wards? What patterns exist in grievance status? Which findings can support municipal planning?

1.1 Objectives

- Validate and combine BBMP grievance records for 2020–2024.
- Clean and standardise dates, categories, wards, statuses, and identifiers.
- Analyse trends, category demand, ward concentration, and status distribution.
- Create KPI outputs and evidence-based decision recommendations.
- Segment wards with K-Means only where the analytical use is justified.
- Build an interactive dashboard for filtered civic intelligence.

1.2 Dataset Scope

The final combined dataset contains 639,674 records, 198 unique wards, 32 categories, and 151 subcategories. The source files cover calendar years 2020, 2021, 2022, 2023, and 2024.

CHAPTER 2

TOOLS & TECHNOLOGY USED

Python was used as the primary implementation language. Pandas supported loading, cleaning, validation, and aggregation. NumPy supported numerical operations. Matplotlib and Seaborn supported static analytical visualisations. Scikit-learn supported feature scaling

and K-Means clustering. Streamlit and Plotly were used to develop the interactive dashboard. Python-docx was used to produce this report.

2.1 Data Processing Workflow

The workflow consists of loading the annual files, validating the expected schema, combining the records, checking duplicate Complaint IDs, parsing Grievance Date, standardising text fields, deriving Year, Month, Quarter, and DayOfWeek, and exporting a reproducible cleaned dataset.

2.2 Machine Learning Methodology

Ward-level civic profiles were constructed using grievance volume, category diversity, status proportions, and proportions of leading categories. The features were standardised using StandardScaler. Candidate K values from 2 through 6 were compared using silhouette score, and the best-performing K was selected for the final K-Means model. Cluster meanings were interpreted only after examining the resulting ward profiles.

Table 1. KPI Summary of CivicPulse

Metric	Value
Total grievances	639,674
Unique wards	198
Unique categories	32
Unique subcategories	151
Top category	Electrical
Top ward	Jnanabharathi Ward

CHAPTER 3

SNAPSHOTS

The following figures are generated from the actual CivicPulse pipeline outputs. They show the annual complaint trend, category distribution, ward demand concentration, and grievance status distribution.

Figure 1. Grievance Volume by Year

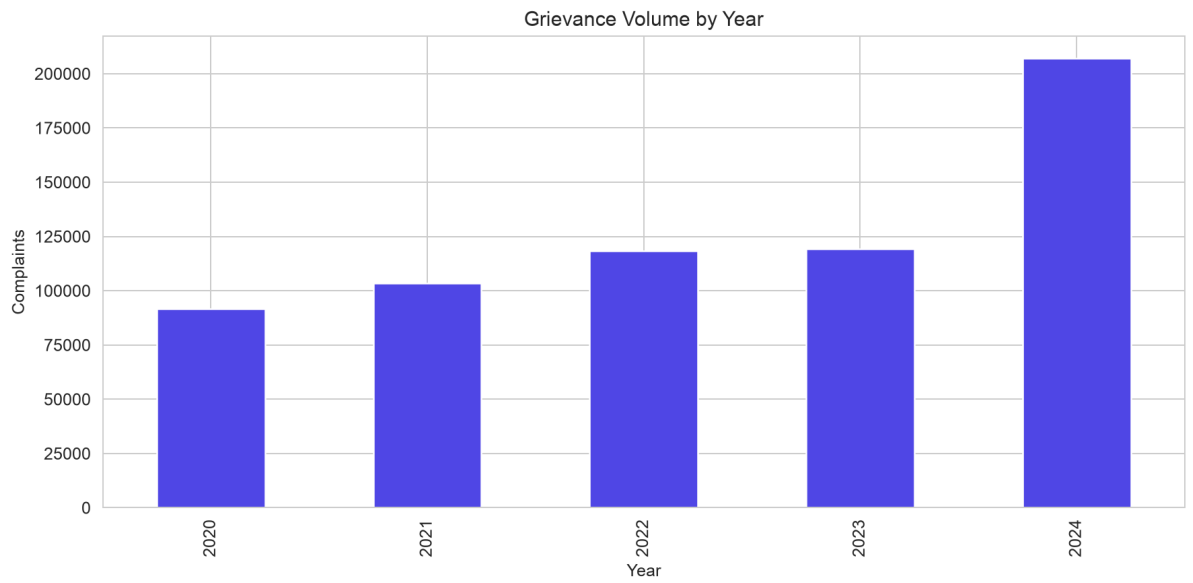


Figure 2. Top Complaint Categories

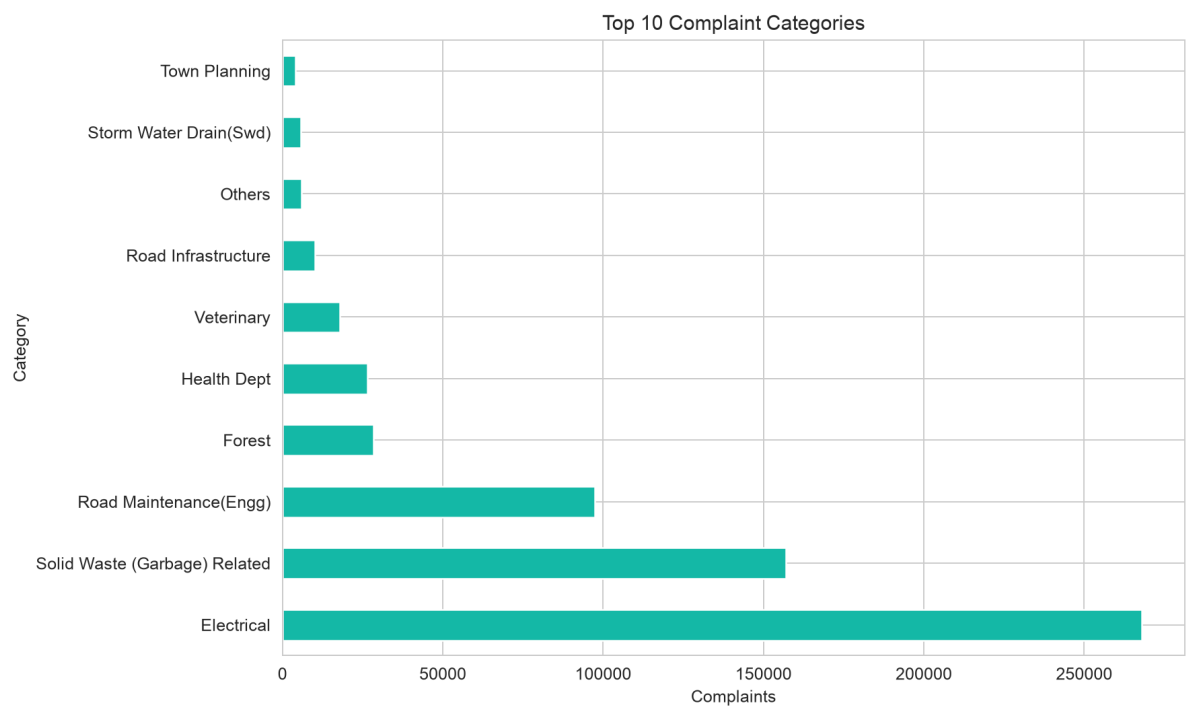


Figure 3. Highest-Demand Wards

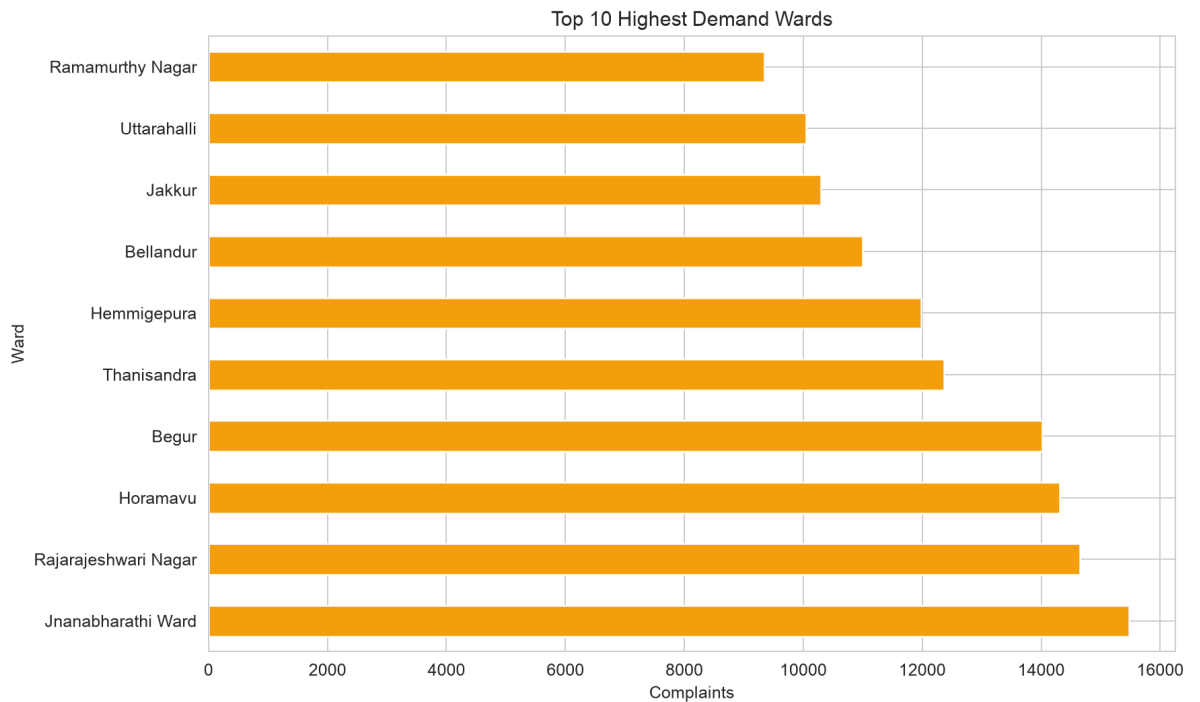
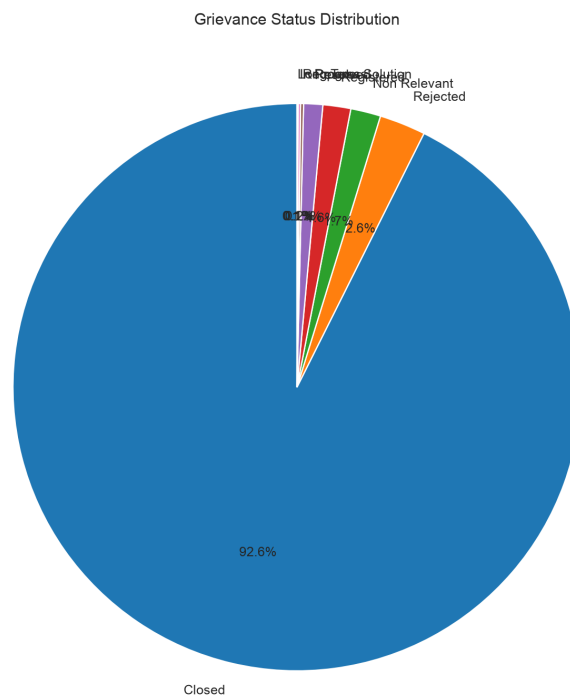


Figure 4. Grievance Status Distribution



CHAPTER 4

RESULTS AND DISCUSSIONS

The combined dataset contains 639,674 valid grievance records. The annual distribution is 91,620 records in 2020, 103,504 in 2021, 118,394 in 2022, 119,140 in 2023, and 207,016 in 2024. The 2024 dataset has the highest recorded volume in the available period.

The dominant category is Electrical, with 267,990 records. It is followed by Solid Waste (Garbage) Related with 157,002 records and Road Maintenance(Engg) with 97,411 records. These categories represent important public-facing service areas and should be treated as priority domains for monitoring and resource planning.

The highest-demand ward is Jnanabharathi Ward, with 15,481 complaints. Other high-demand wards include Rajarajeshwari Nagar, Horamavu, Begur, and Thanisandra. This uneven concentration supports a ward-level rather than uniform citywide intervention strategy.

Status analysis shows that Closed is the most common recorded status, followed by Rejected, Non Relevant, Registered, Resolved, ReOpen, In Progress, and Long Term Solution. Status labels are retained as published rather than being renamed into unsupported categories such as “Critical” or “Pending”.

The ward segmentation component provides a descriptive grouping of wards with similar complaint intensity, diversity, and status profiles. It is intended to support prioritisation and comparative planning, not to claim causality. The model evaluation and cluster summaries are stored in the generated project outputs.

Table 2. Top Complaint Categories

Rank	Category	Complaints
1	Electrical	267,990
2	Solid Waste (Garbage) Related	157,002
3	Road Maintenance(Engg)	97,411
4	Forest	28,293
5	Health Dept	26,546

Table 3. Highest-Demand Wards

Rank	Ward	Complaints
1	Jnanabharathi Ward	15,481
2	Rajarajeshwari Nagar	14,646
3	Horamavu	14,309
4	Begur	14,015
5	Thanisandra	12,366

Table 4. Grievance Status Distribution

Status	Complaints
Closed	592,468

Status	Complaints
Rejected	16,803
Non Relevant	10,932
Registered	10,147
Resolved	6,953
ReOpen	1,043
In Progress	896
Long Term Solution	412

4.1 Decision Implications

- Prioritise electrical, solid waste, and road maintenance categories for recurring service monitoring.
- Use ward-level demand rankings to support targeted deployment and preventive maintenance.
- Monitor registered, reopened, in-progress, and long-term-solution cases separately from closed records.
- Use seasonal and monthly views in the dashboard to plan operational capacity.
- Treat cluster membership as a comparative planning aid and validate decisions with departmental context.

CHAPTER 5

CONCLUSIONS AND FUTURE SCOPE

CivicPulse demonstrates how historical civic grievance records can be converted into a reproducible analytical decision-support project. The analysis identifies the largest complaint categories, the wards with the greatest recorded demand, the distribution of operational statuses, and the increasing volume of recorded complaints across the study period.

The main conclusion is that civic demand is unevenly distributed by both category and ward. A small number of service areas account for a large portion of complaints, while a group of wards experiences higher recorded demand than others. This supports targeted ward-wise planning and category-specific preventive action.

Future work may include validated monthly forecasting, integration with ward population and infrastructure indicators, before-and-after evaluation of municipal interventions, geospatial mapping, and richer dashboard access controls. Forecasting should be added only after confirming sufficient historical granularity and an appropriate validation strategy.

REFERENCES

BBMP grievance data files provided for the CivicPulse internship project, 2020–2024.

AICTE – IBM SkillsBuild Data Analytics with AI Internship Program 2026 materials.

BharatCares internship project context and workflow documentation.

Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Streamlit, Plotly, and python-docx documentation.

OpenCity civic data context for municipal service analytics.

DATA SHEET

The project data sheet is supplied as the cleaned CSV file `data/processed/bbmp_grievances_clean.csv`. The source schema contains Complaint ID, Category, Sub Category, Grievance Date, Ward Name, Grievance Status, Staff Remarks, and Staff Name. Derived fields are Year, Month, Quarter, and DayOfWeek.

APPENDICES

Appendix A: `DATA_DICTIONARY.md`

Appendix B: `data_audit_report.md`

Appendix C: Generated figures in `outputs/figures/`

Appendix D: CivicPulse source code and dashboard instructions in `README.md`